

API INTERVIEW QUESTIONS (100)

Beginner + Advanced + Real-Time Scenarios

SECTION 1: API BASICS (Q1-Q30)

1. What is an API?

Definition: An API (Application Programming Interface) is a set of rules and protocols that allows different software applications to communicate with each other.

Real-time Scenario:

When you use a weather app on your phone, it doesn't have weather data built-in. Instead, it calls a weather service API (like Open WeatherMap), sends your location, and receives current weather data to display.

2. What is REST API?

Definition: REST (Representational State Transfer) is an architectural style for designing networked applications that use HTTP methods to perform operations on resources.

Real-time Scenario:

A food delivery app uses REST APIs:

- GET /restaurants - Fetch available restaurants

- POST /orders - Place a new order

- GET /orders/(id) - Check order status

- DELETE /orders/(id) - Cancel an order

3. What are HTTP methods?

Definition: Standard operations defined by HTTP protocol for web communication.

Real-time Usage:

- GET /api/products

- # Get all products

- (E-commerce site)

- POST /api/cart

- # Add item to cart

- PUT /api/users/123

- # Update user profile

- DELETE /api/comments/5 # Remove comment

- PATCH /api/orders/10

- # Update order status partially

4. Difference between GET and POST?

GET:

- Used to retrieve data

- Parameters visible in URL

- Bookmarkable

- Limited data capacity

- Example: Searching products on Amazon: amazon.com/search?q=laptop

POST:

- Used to submit data

- Data in request body (hidden)

- Not bookmarkable

- Larger data capacity

- Example: Submitting login form with username/password

5. What is an endpoint?

Definition: A specific URL where an API can be accessed to perform operations.

Real-time Example:

Banking Application:

- /api/accounts

- POST

- /api/accounts/(id)

- /api/transfers /api/transactions

- List all accounts

→ Get specific account
Initiate money transfer
View transaction history

6. What is request and response?

Request: What client sends to server

POST /api/login HTTP/1.1

Content-Type: application/json ("username": "john",
"password": "pass123")

Response: What server returns to client

HTTP/1.1 200 OK

Content-Type: application/json ("token": "abc123",
"user": {"id":

1,
"name": "John" }}

7. What is JSON?

Definition: JSON (JavaScript Object Notation) is a lightweight data interchange format that's easy for humans to read and write, and easy for machines to parse.

Real-time Scenario:

When you fill out a registration form, your data is converted to JSON:

"firstName": "Sarah",
"lastName": "Chen",
"email": "sarah@email.com",
"phone": "+1234567890",
"address": {
"street": "123 Main St",
"city":
"San Francisco",
"zipcode": "94107"

8. What is HTTP status code?

Definition: A 3-digit code that indicates the result of an HTTP request.

Real-time Examples:

200 OK - Your request succeeded (Google search results)

404 Not Found - Page doesn't exist (broken link)

500 Internal Server Error - Website is down

403 Forbidden - You're not allowed to access
(admin panel without login)

9. Common HTTP status codes?

Success:

200 OK - Request successful

201 Created - Resource created successfully (new user registered)

204 No Content - Success but no data to return

Client Errors:

• 400 Bad Request - Invalid input (wrong email format)

401 Unauthorized - Not authenticated (missing token)

403 Forbidden - Authenticated but not authorized (user trying to access admin area)

404 Not Found - Resource doesn't exist

429 Too Many Requests - Rate limit exceeded

Server Errors:

500 Internal Server Error - Server problem

502 Bad Gateway - Upstream server issue

503 Service Unavailable - Server overloaded/maintenance

10. What is RESTful API?

Definition: An API that follows REST architectural principles including statelessness, cacheability, and proper use of HTTP methods.

Real-time Scenario - E-commerce API:

GET

/products

POST

/products

GET

/products/ (id)

PUT

/products/(id)

PATCH /products/(id)

DELETE /products/(id)

→ Returns list of products

→ Create new product (admin)

Get specific product details

Update entire product

Update product price only

→ Remove product

11. What is statelessness?

Definition: Each request from client to server must contain all the information needed to understand and process the request. Server doesn't store any client context between requests.

Real-time Example:

When you browse products on Amazon:

- Request 1: GET /products
- Request 2: GET /cart (token required)
- Request 3: GET /products/123

12. What is CRUD?

Definition: CRUD stands for Create, Read, Update, Delete.

Real-time Blog System:

Create:

POST /articles

Read:

GET /articles/{id}

Update: PUT /articles/{id}

Delete:

DELETE /articles/{id}

13. What is API testing?

Definition: Verifying API functionality, performance, security, and reliability.

Real-time Scenario:

Testing payment APIs for success, failure, load, and edge cases.

14. Tools for API testing?

Postman

Swagger/OpenAPI

SoapUI

15. What is payload?

Definition: Data sent in request or response body.

Example:

"content": "Just launched my new app!",

"privacy":

"public"

}

Perfect 3

I'll clearly explain each question (16-30) with:

Practical usage

No shortcuts, no confusion — this is exactly what interviewers expect.

16. What are headers?

Definition

Headers are key-value pairs sent with an HTTP request or response that contain metadata (extra information) about the request or response.

They do not contain actual data, but they tell the server how to process the data.

Real-time Example

When a user opens a dashboard after login:

GET /api/user/profile

Authorization: Bearer eyJhbGciOiJIUzIiNi...

Content-Type: application/json

Accept: application/json

Authorization → Who the user is

Content-Type → Format of request data

Accept → Expected response format

Interview Tip

- Headers are mainly used for authentication, content type, caching, and client information.

17. What is Content-Type?

Definition

Content-Type header specifies the format of the data being sent in the request or response body.

Real-time Examples

1. Sending JSON data

Content-Type: application/json

"email": "user@gmail.com",

"password":

"123456"

2. File upload

Content-Type: multipart/form-data

Why it is important

If the server receives JSON but Content-Type is missing or wrong, the request may fail or throw errors.

Interview Tip

- Always set Content-Type correctly, especially for POST, PUT, PATCH requests.

18. What is Accept header?

Definition

The Accept header tells the server which response format the client expects.

Real-time Example

Mobile App Request

GET /api/products

Accept: application/json

Server responds:

"id": 101,

"name": "Laptop",

"price": 50000

Browser Request

Accept: text/html

Server responds with an HTML page.